Team Details

a. **Team name:** **APEX**
b. **Team leader name:** **Karan Shedge**
- **Problem Statement:** **Smart Campus Navigator: AI-Powered Wayfinding Revolution**

Brief about your solution and problem statement addressing

Problem Statement

Navigating a large college campus can be confusing for students, visitors, and new staff, especially when trying to locate specific wings, facilities, or rooms on the ground floor. The absence of an easy-to-use, interactive navigation system often leads to wasted time, wrong directions, and dependence on others for guidance.

Solution

The project is a college campus map navigation system focused on the ground floor layout. The solution provides a digital map of the campus where users can mark their starting point and destination. Based on the selected points, the map helps users easily understand how to reach key locations such as different wings (R, L, O, S, T, P, J, and other facilities shown in the campus layout, G Wings), canteen, workshop, architecture labs, toilets, parking, court,

This system improves accessibility and orientation by offering a visual, user-friendly guide, making campus navigation faster, simpler, and more efficient—especially for newcomers and visitors.

Opportunities

Difference from existing ideas:

- Traditional signboards and fixed layouts provide limited guidance, whereas this project delivers a dynamic ground-floor campus guide for easier orientation.
- By allowing users to select their current location and desired place, the system ensures smooth, quick, and confusion-free navigation within the campus.

Problem-solving approach:

- Offers a well-organised ground-floor layout that helps users understand campus structure before moving.
- Functions as a self-guided navigation aid, reducing dependence on help desks or others.
- Improves time management and user confidence, especially for first-time visitors.

List of features offered by the solution

- Landmark-based navigation using easily recognisable campus spots
- Zoom and pan functionality for better map clarity
- Colour-coded wings and facilities for faster identification
- Search option for quick location finding
- Visitor-friendly mode for first-time users
- Simple and lightweight design for smooth performance
- Future-ready architecture for integration with smart campus systems

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Google Technologies used in the solution

- 1. Firebase Authentication 🔐
- 2. Cloud Firestore Database ☁
- 3. Firebase SDK Integration□

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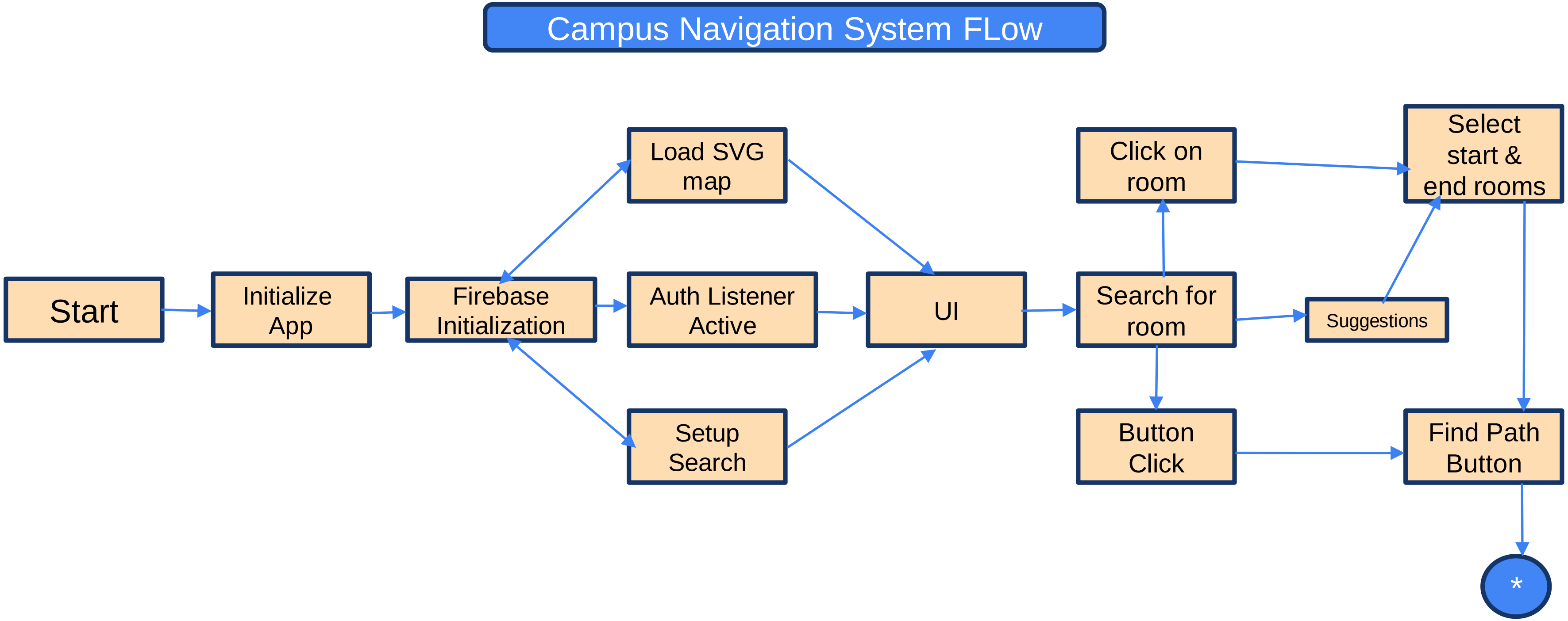
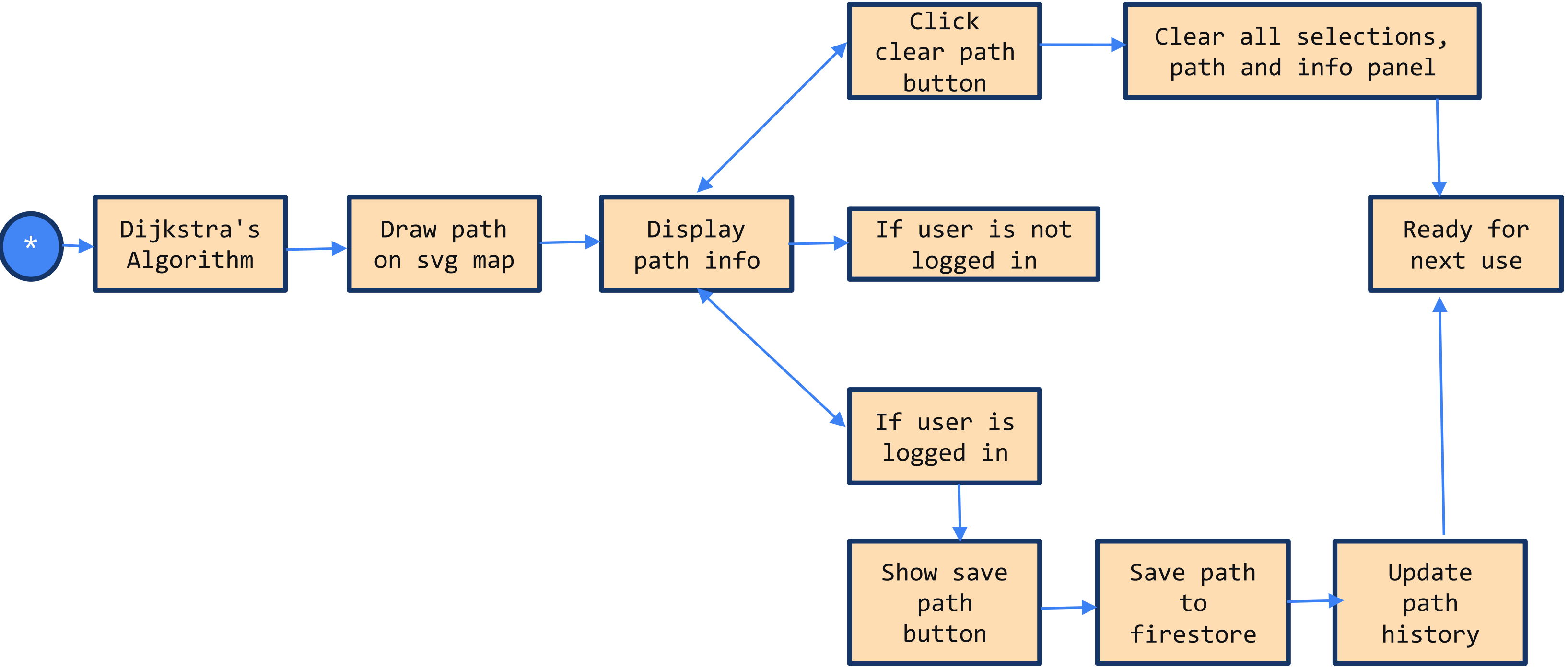
Process flow diagram



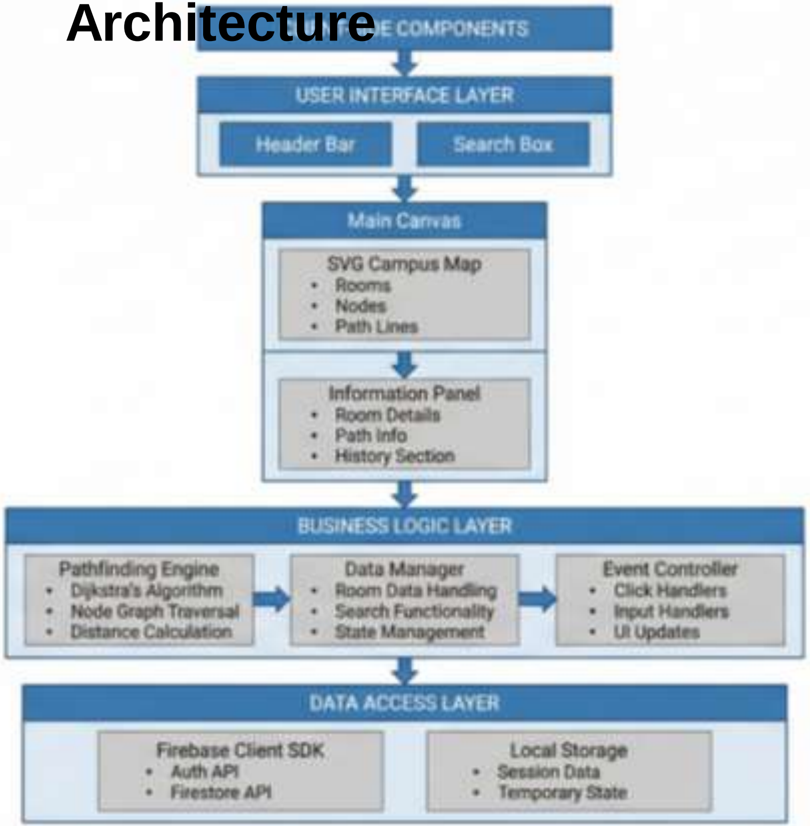

*
Dijkstra's Algorithm
Draw path on svg map
Display path info
Click clear path button
Clear all selections, path and info panel
If user is not logged in
Ready for next use
If user is logged in
Show save path button
Save path to firestore
Update path history

Architecture diagram of the proposed solution

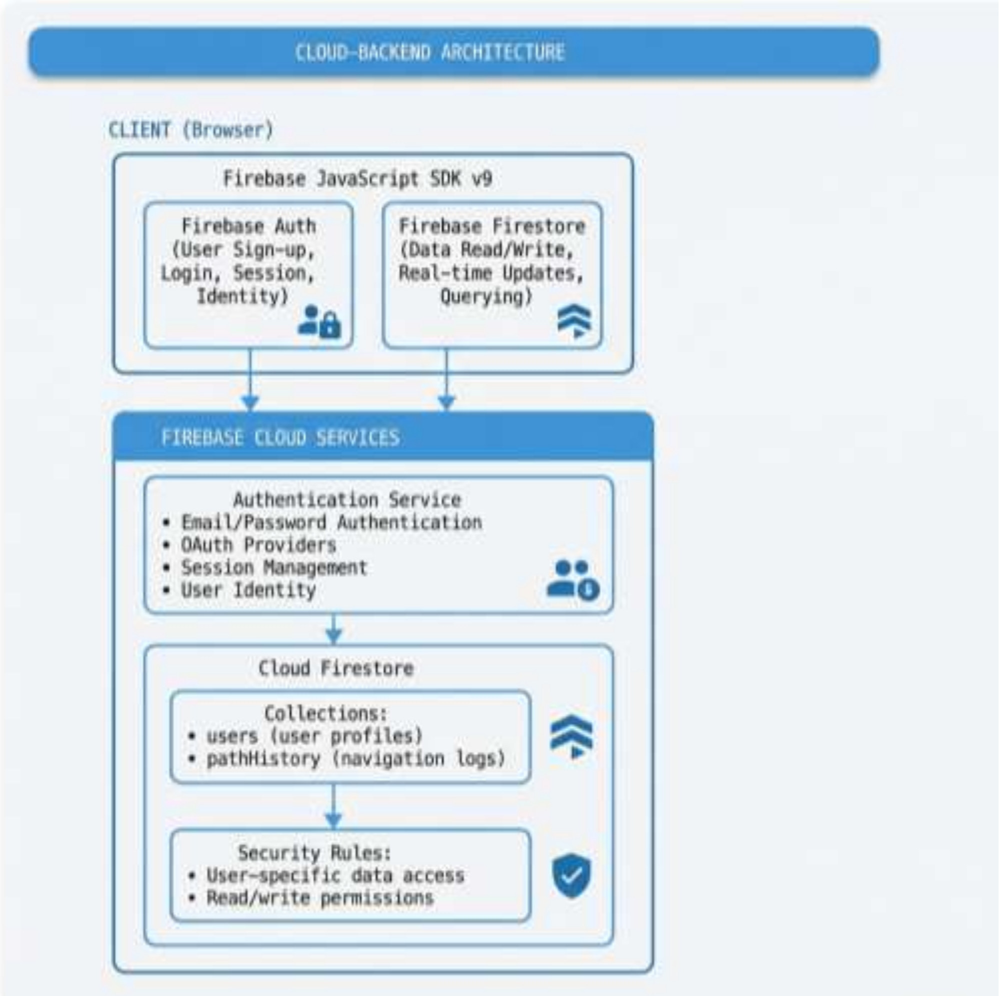
1. HIGH-LEVEL SYSTEM ARCHITECTURE

```text
THREE-TIER ARCHITECTURE

PRESENTATION LAYER
(Client-Side / Frontend)

• HTML5/CSS3/JavaScript
• SVG Vector Campus Map
• Responsive UI Components
• Real-time User Interactions

  | HTTP/HTTPS
  ▼
APPLICATION LAYER
(Business Logic)

• Dijkstra's Pathfinding Algorithm
• Room Search & Filtering
• User Session Management
• Data Validation & Processing

  | API Calls
  ▼
DATA LAYER
(Persistence)

• Firebase Authentication
• Cloud Firestore Database
• Local Browser Storage
• Static Campus Data (Rooms/Nodes)
```

Detailed Component Architecture



3. FIREBASE INTEGRATION ARCHITECTURE



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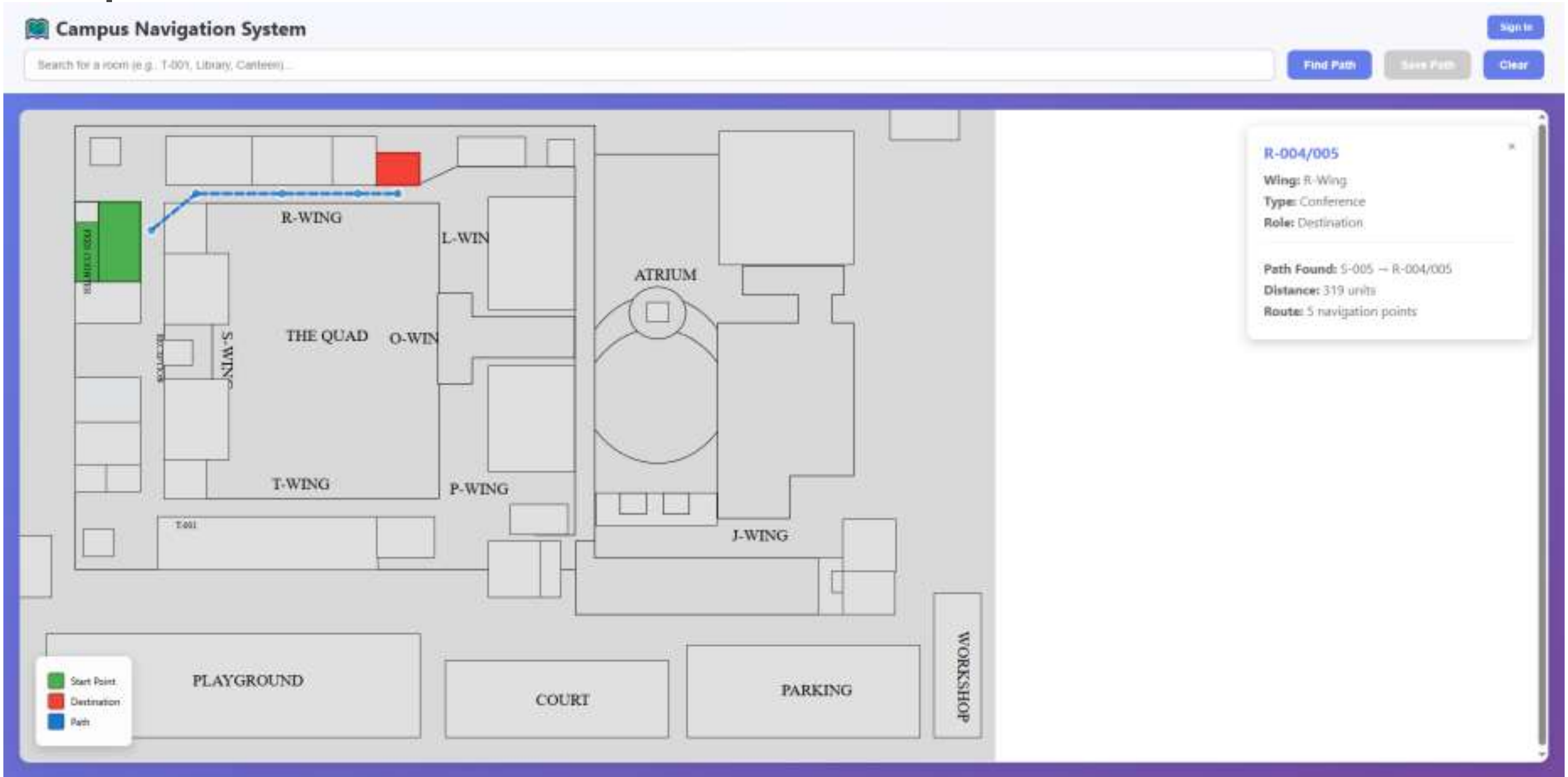
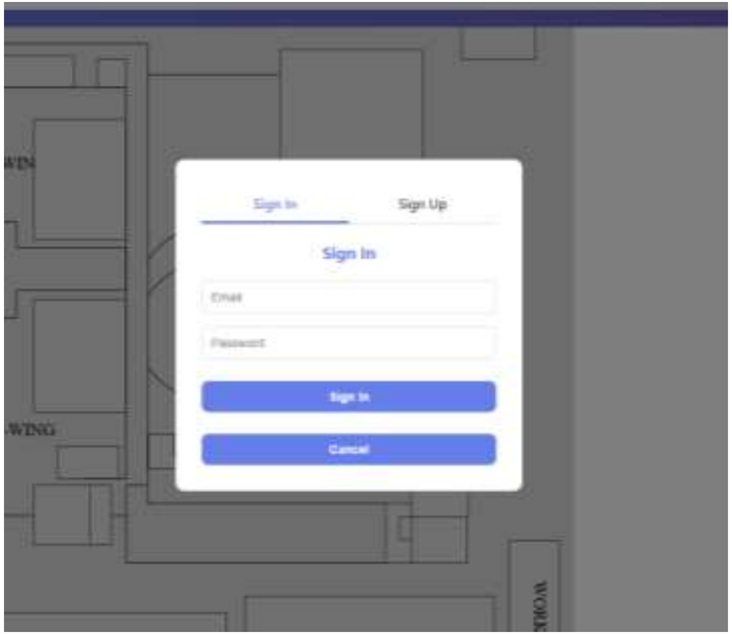
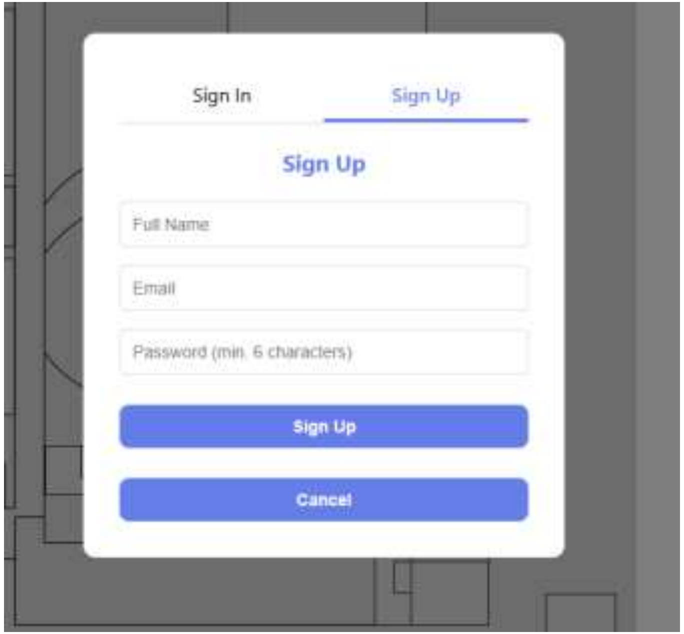
Snapshots of the MVP




Sign In
Sign Up
Sign In
Email
Password
Sign In
Cancel


Sign In
Sign Up
Sign Up
Full Name
Email
Password (min. 6 characters)
Sign Up
Cancel

Additional Details/Future Development (if any)

Development:

- ➢ Expansion to cover all floors and campus buildings
- ➢ Integration of real-time indoor navigation and voice guidance
- ➢ Development of a mobile-friendly or Android application
- ➢ Inclusion of accessibility features and multilingual support

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Provide links to your:

1. **GitHub Public Repository :** https://github.com/Kayyy15/GDG-Techsprint-AI-Hack-25
2. **Demo Video Link (3 Minutes)**
3. **MVP Link**

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Leveraging the power of AI

Thank you!